Test Table Including Test Results

Test No.	Test Purpose	Test Data Type (Normal, Extreme, Erroneous Where Required) If Any Validation Rules Are Set ?	Expected Results	Actual Results	PASS/FAIL & Comments
1	Test BMI with lower extreme data input	Height = 120CM, Weight = 40KG	Calculated 27.78 Weight ÷ Height m^2 (Patient,2024)	28	PASS due to BMI results being rounded to the nearest whole number, 27.78 is rounded up
2	Test BMI with normal data input	Height = 179CM , Weight = 77KG	Calculated 24.03 Weight ÷ Height m^2	24	PASS
3	Test BMI with upper extreme data input	Height = 200CM, Weight = 250KG	Calculated 62.50 Weight ÷ Height m^2	62	PASS
4	Test BMR Harris-Benedict calculation for males with lower extreme data input	Age = 18 YEARS, Height = 120CM. Weight = 40KG	Calculated 1095.4 66.5 + (13.75 x Weight) + (5.003 x Height) – (6.75 x Age)	1095	PASS due to BMI results being rounded to the nearest whole number, 1095.4 is rounded down
5	Test BMR Harris-Benedict calculation for males with normal data input	Age = 28 YEARS, Height = 180CM. Weight = 77KG	Calculated 1836.8 66.5 + (13.75 x Weight) + (5.003 x Height) – (6.75 x Age)	1836	PASS
6	Test BMR Harris-Benedict calculation for male with upper extreme data input	Age = 100 YEARS, Height = 200CM. Weight = 250KG	Calculated 3829.6 66.5 + (13.75 x Weight) + (5.003 x Height) – (6.75 x Age)	3827	PASS Abnormal data the result has a discrepancy of 3 & should been rounded up to 3830
7	Test BMR Harris-Benedict calculation for females with lower extreme data input	Age = 18 YEARS, Height = 120CM. Weight = 40KG	Calculated 1212.5 655.1 + (9.563 x Weight) + (1.850 x Height) – (4.676 x Age)	1175	PASS Abnormal data the result has a discrepancy of 38 & should been rounded up 1213
8	Test BMR Harris-Benedict calculation for females with normal data input	Age = 28 YEARS, Height = 151CM. Weight = 55KG	Calculated 1329.5	1329	PASS

			655.1 + (9.563 x Weight) + (1.850 x Height) – (4.676 x Age)		
9	Test BMR Harris-Benedict calculation for females with lower extreme data input	Age = 100 Years, Height = 200CM, Weight = 250KG	Calculated 2948.3 655.1 + (9.563 x Weight) + (1.850 x Height) – (4.676 x Age)	2948	PASS
10	Test BMR Mifflin-St. Jeor calculation for males with lower extreme data input	Age = 18 YEARS, Height = 140CM, Weight = 40KG	Calculated 1065 10 x Weight + 6.25 x Height – 5 x Age + 5	1065	PASS The Mifflin-St Jeor equation also seem round the BMR result
11	Test BMR Mifflin-St. Jeor calculation for males with normal data input	Age = 25 YEARS, Height = 180CM, Weight = 77KG	Calculated 1775 10 x Weight + 6.25 x Height – 5 x Age + 5	1775	PASS
12	Test BMR Mifflin-St. Jeor calculation for males with upper extreme data input	Age = 100 YEARS, Height = 200CM, Weight = 250KG	Calculated 3255 10 x Weight + 6.25 x Height – 5 x Age + 5	3255	PASS
13	Test BMR Mifflin-St. Jeor calculation for females with lower extreme data input	Age = 18 YEARS, Height = 120CM, Weight = 40KG	Calculated 899 10 x Weight + 6.25 x Height – 5 x Age - 161	899	PASS Bug in system displays the result overlapped where another number is visible making the result read as 8995
14	Test BMR Mifflin-St. Jeor calculation for females with normal data input	Age = 25 YEARS, Height = 147CM, Weight = 50KG	Calculated 1133 10 x Weight + 6.25 x Height – 5 x Age - 161	1113	PASS
15	Test BMR Mifflin-St. Jeor calculation for females with upper extreme data input	Age = 100 YEARS, Height = 200CM, Weight = 250KG	Calculated 3089 10 x Weight + 6.25 x Height – 5 x Age - 161	3089	PASS

16	Test Colour Chooser option is functional	Select “Black” from the colour palette & confirm (OK) selection, repeat this process with “RED” & “Green” as the selected colours. Validation gained via set colour selection palette, so erroneous data cannot be entered.	1. Default background colour should change to black. 2. black background colour should change to Red. 3. Red background colour should change to Green.	1. Default background colour changed to Black. 2. Black background colour changed to Red. 3. Red background colour changed to Green.	PASS
17	Test Exercise Routine chosen is “None” which should multiple the BMR Harris-Benedict calculation result for males by 1.2 .	Age = 25 YEARS, Height = 180CM, Weight = 77KG, Exercise Routine = NONE Validation gained via set combo box (dropdown menu) , so erroneous data cannot be entered.	Calculated 2228 66.5 + (13.75 x Weight) + (5.003 x Height) – (6.75 x Age) x 1.2	BMR with exercise is 2227. Due the BMR Calculation being rounded before the exercise multiplication.	PASS

18	Test Exercise Routine chosen is "Light (1-3 Days/Week)" which should multiple the BMR Mifflin-St Jeor calculation result or males by 1.375.	Age = 25 YEARS, Height = 180CM, Weight = 77KG, Exercise Routine = LIGHT (1-3 DAYS-WEEK)	Calculated 2441 10 x Weight + 6.25 x Height – 5 x Age + 5 x 1.375	BMR with exercise is 2441.	PASS
19	Test Exercise Routine chosen is "Moderate (3-5 Days/Week)" which should multiple the BMR Harris-Benedict calculation result for females by 1.55 .	Age = 25 YEARS, Height = 150CM, Weight = 77KG, Exercise Routine = MODERATE (3-5 DAYS/WEEK)	Calculated 2406 66.5 + (13.75 x Weight) + (1.850 x Height) – (4.676 x Age) x 1.55	BMR with exercise is 2406.	PASS
20	Test Exercise Routine chosen is "Heavy (5-6 Days/Week)" which should multiple the BMR Mifflin-St Jeor calculation result for females by 1.725.	Age = 25 YEARS, Height = 150CM, Weight = 77KG, Exercise Routine = HEAVY (5-6 DAYS/WEEK)	Calculated 2452 10 x Weight + 6.25 x Height – 5 x Age – 161 x 1.725	BMR with exercise is 2453.	PASS
21	Test Exercise Routine chosen is "Intense (7 Days/Week)" which should multiple the BMR Harris-Benedict calculation result for males by 1.9	Age = 25 YEARS, Height = 180CM, Weight = 77KG, Exercise Routine = INTENSE (7 DAYS/WEEKS)	BMR with Exercise = 3528 66.5 + (13.75 x Weight) + (5.003 x Height) – (6.75 x Age) x 1.9	BMR with exercise is 3526.	PASS
22	Test BMI result displays weight category as "UNDERWEIGHT"	Height = 200CM, Weight = 40KG	BMI = 10.00 , Weight Category = UNDERWEIGHT	BMI = 10 , Weight Category = UNDERWEIGHT	PASS
23	Test BMI result displays weight category as "HEALTHY WEIGHT"	Height = 180CM Weight = 70KG	BMI = 21.60 , Weight Category = HEALTHY WEIGHT	BMI = 22 , Weight Category = HEALTHY WEIGHT	PASS

24	Test BMI result displays weight category as "OVERWEIGHT". lower extreme data input was used	Height = 120CM Weight = 40KG	BMI = 27.78 , Weight Category = OVERWEIGHT	BMI = 28 , Weight Category = OVERWEIGHT	PASS
25	Test BMI result displays weight category as "OBESE". upper extreme data input was used	Height = 200CM Weight = 250KG	BMI = 62.50 , Weight Category = OBESE	BMI = 62 , Weight Category = OBESE	PASS
26	Test the application installs & runs independently	Loaded the installation file on a USB flash drive, plug the USB into another computer to install & run	the application should install. Creates a desktop icon & runs	The application installed & runs	PASS The application EXE was packaged into a Microsoft Software Installer file for ease of installation purposes.

Test Results – Conclusion & Comments on Robustness of Application

The expected test results for the calculations were conduct using official online BMR (with relevant formulas) & BMI calculators. The actual application test results had some abnormal & discrepancies in the data, this was re-examined with a scientific calculator which confirmed the actual results were accurate. My findings are that some online calculators have different interpretation of data; this may be due to other factors being included or errors in the calculations. All the major functions of the developed application work correctly. In conclusion the application is fit for its purpose. The application is robust in term data entry no margin for errorless data, , can perform number of complex calculations with a range of data input in variety combination & output accurate results.